

**A PROJECT TITLE
ON**

Explainable Hybrid Machine Learning Framework for Predictive Maintenance
project report submitted in partial fulfilment of the requirements for the Course

**MACHINE LEARNING
BACHELOR OF TECHNOLOGY**

**In
A.Y.: 2026-27**

By

Name: V. Navadeep Roll: 2520030019

Name: K. Shankar Roll: 2520030063

**Under the Esteemed guidance of
Dr.N.Nirmalajyothi
Associate Professor, CSE**

Department of Computer Science and Engineering



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Abstract

This project presents the design, implementation, and evaluation of an **explainable hybrid machine learning framework for predictive maintenance of industrial machinery** using the **AI4I 2020 Predictive Maintenance Dataset**. The framework uses machine operating parameters such as air temperature, process temperature, rotational speed, torque, tool wear, and machine type to predict machine failures. The proposed methodology includes data preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and explainability. Derived features such as **temperature difference** and a **mechanical power proxy** are generated to capture additional operating characteristics. Multiple machine learning classifiers, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, are evaluated. The proposed hybrid model combines the predicted probabilities of **Random Forest and XGBoost** to generate the final machine-failure prediction. Since failure instances are relatively rare, model performance is evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Precision-Recall AUC rather than relying only on accuracy. To improve model transparency, **SHAP (SHapley Additive exPlanations)** is applied to identify the features that contribute most strongly to failure predictions. The framework therefore provides both machine-failure prediction and interpretable explanations, supporting more informed and trustworthy predictive-maintenance decisions.

Keywords: Predictive Maintenance, Machine Learning, Hybrid Machine Learning, Random Forest, XGBoost, Explainable Artificial Intelligence (XAI), SHAP, AI4I 2020.

1. Introduction

1.1 Purpose of the Project

Unplanned machine failure is one of the costliest disruptions faced by manufacturing plants, power stations, and process industries, leading to lost production, safety hazards, and expensive emergency repairs. Traditional maintenance strategies, reactive maintenance carried out after failure and scheduled preventive maintenance carried out at fixed intervals, either waste healthy component life or fail to catch faults before they occur. Predictive maintenance addresses this gap by continuously monitoring sensor signals such as vibration, temperature, pressure, and current draw, and using machine learning to forecast when a failure is likely, allowing maintenance to be scheduled only when it is actually needed.

The stakeholders of such a system include plant reliability engineers, maintenance technicians, production managers, and safety officers, all of whom need not only an accurate failure alarm but also a clear explanation of why the model raised that alarm before they will act on it or trust it. Purely black-box machine learning models, however accurate, are often rejected on the shop floor because operators cannot verify their reasoning. This motivates an explainable hybrid ML solution that couples strong predictive accuracy with transparent, feature-level explanations, so that predictions can be audited, trusted, and acted upon with confidence.

1.2 Problem Statement

Given time-stamped multivariate sensor readings, such as vibration amplitude, bearing temperature, motor current, pressure, torque, and rotational speed, collected from industrial machines, the problem is framed as a classification task that predicts whether a machine will fail within a defined future time window, or which specific fault type is developing, and, for every prediction, generates a ranked, human-readable explanation of the sensor

features that contributed most to that specific prediction, so that the output functions both as an early-warning signal and as an interpretable justification for the maintenance team.

1.3 Existing Methods and Disadvantages

Existing predictive-maintenance approaches range from physics-based degradation models to purely data-driven machine learning classifiers such as Random Forests, Support Vector Machines, and deep neural networks. Physics-based models require detailed knowledge of machine dynamics and often generalize poorly across different equipment types, while standalone statistical classifiers frequently struggle with the highly imbalanced nature of failure data, since failures are rare events, leading to poor recall on the minority failure class and misleadingly high overall accuracy scores.

More critically, high-performing deep learning and ensemble models are typically opaque black boxes: they output a failure probability without indicating which sensor readings or operating conditions drove that prediction. This lack of interpretability makes it difficult for maintenance engineers to validate alarms, diagnose the underlying fault, or build trust in the system, and it also hides cases of overfitting to spurious correlations, sensor noise, or data leakage introduced by improper train/test splitting of time-series data. These limitations, namely poor generalization, class-imbalance bias, and lack of transparency, motivate the need for a hybrid model that is both accurate and explainable.

1.4 Proposed System

The proposed system uses a **hybrid machine learning approach** for predictive maintenance. After data preprocessing and feature engineering, **Random Forest and XGBoost** are trained to predict machine failure. Their prediction probabilities are combined to generate the final prediction. **SHAP** is then used to explain the important features contributing to each prediction. This provides both **accurate failure prediction and understandable explanations** for maintenance decisions.

1.5 Research Objectives(atleast 4)

1. Collect and prepare a labeled multivariate sensor dataset suitable for predictive-maintenance modeling.
2. Perform systematic preprocessing, missing-value treatment, and exploratory analysis of the sensor data.
3. Engineer and select relevant time-domain features while preventing data leakage across time.
4. Develop and train a hybrid machine learning model that combines tree-based and sequence-based learners for failure prediction.
5. Evaluate and compare the hybrid model against baseline classifiers using accuracy, precision, recall, F1-score, and ROC-AUC.
6. Integrate SHAP and LIME explainability into the best-performing model and interpret feature contributions for practical maintenance decisions.

2. Literature Survey (recent studies from 2-3 years)

Review ten recent and directly relevant studies. Verify bibliographic details, DOI, volume, issue, pages/article number, and year before final submission.

S.No	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
1	SHAP-Driven Explainability in Survival Analysis for Predictive Maintenance	Kargar-Sharif-Abad, M., Kharazian, Z., Miliou, I., & Lindgren, T. (2024). ECAI 2024 Workshop on Human-Aware AI in Industry 5.0 (HAI5.0).	Survival-based RUL models give a risk score but no feature-level	Apply SHAP to survival-analysis based predictive-maintenance	Survival analysis models combined with SHAP value	SHAP explanations for survival models reveal that different	Extend SHAP-based survival explainability to multi-sensor industrial datasets beyond the

	Applications		reasoning for why a machine is flagged as high risk.	models to explain individual risk predictions.	attribution for feature-level explanation.	sensor features drive short-term versus long-term failure risk.	benchmark case studied.
2	Evaluating eXplainable Artificial Intelligence Tools for Hard Disk Drive Predictive Maintenance	Ferraro, A., Galli, A., Moscato, V., & Sperli, G. (2023). Artificial Intelligence Review, 56(7), 7279-7314, doi:10.1007/s10462-022-10354-7.	Multiple XAI tools such as SHAP and LIME give inconsistent explanations for the same failure model, making tool choice unclear.	Systematically compare several explainability tools on a real hard-disk failure dataset.	Comparative evaluation of SHAP, LIME, and other post-hoc XAI methods on gradient-boosted failure classifiers.	Explanation quality and consistency vary substantially across XAI tools, so a single tool cannot be assumed reliable for all models.	Combine multiple explanation methods with quantitative explanation-quality metrics before deploying an XAI tool in production.
3	An Explainable Predictive Maintenance Strategy for Multi-Fault Diagnosis of Rotating Machines Using Multi-Sensor Data Fusion	Gawde, S., Patil, S., Kumar, S., Kamat, P., & Kotecha, K. (2024). Decision Analytics Journal, 10, Article 100425, doi:10.1016/j.dajour.2024.100425.	Single-sensor fault-diagnosis models miss complementary information available across vibration, acoustic, and thermal channels.	Fuse multi-sensor data for multi-fault diagnosis while providing an interpretable explanation for each diagnosis.	Multi-sensor data fusion combined with ensemble classifiers and post-hoc explainability (SHAP/LIME).	Sensor fusion improves multi-fault classification accuracy, and explanations highlight which sensor modality is most informative per fault type.	Investigate real-time sensor-fusion pipelines and edge deployment of the explainable multi-fault model.
4	Explainable Predictive Maintenance: A Survey of Current Methods, Challenges and Opportunities	Cummins, L., Sommers, A., Ramezani, S. B., Mittal, S., Jabour, J., Seale, M., & Rahimi, S. (2024). IEEE Access, 12, 57574-57602, doi:10.1109/ACCESS.2024.3391130.	The predictive-maintenance literature lacks a unified categorization of explainability methods and their suitability for different tasks.	Systematically review and categorize explainable-AI methods applied to predictive maintenance using a PRISMA-based methodology.	Systematic literature review and taxonomy of model-agnostic and model-specific XAI techniques.	Most explainable predictive-maintenance studies rely on post-hoc, model-agnostic methods, and rigorous human-centered evaluation of explanations remains rare.	Develop standardized benchmarks and human-in-the-loop evaluation protocols for explainable predictive-maintenance systems.
5	Prediction and Classification of IoT Sensor Faults Using Hybrid Deep Learning Model	Seba, A. M., Gemed, K. A., & Ramulu, P. J. (2024). Discover Applied Sciences, 6, Article 9, doi:10.1007/s42452-024-05633-7.	Resource-limited IoT sensors produce noisy, fault-prone data that single deep-learning models fail to classify reliably.	Design a hybrid deep-learning model to both predict and classify sensor faults from IoT time-series data.	Hybrid deep learning architecture combining convolutional and recurrent layers for sensor-fault prediction and classification.	The hybrid deep-learning design outperforms single-architecture baselines in detecting and categorizing sensor faults.	Extend the hybrid deep-learning approach as a foundation for broader predictive-maintenance and fault-diagnosis applications.
6	An EWS-LSTM-Based Deep Learning Early Warning System for Industrial Machine Fault Prediction	Cassano, F., Crespino, A. M., Lazoi, M., Specchia, G., & Spennato, A. (2025). Applied Sciences, 15(7), Article 4013, doi:10.3390/app15074013.	Conventional threshold-based alarms give late or false warnings and do not adapt to gradual, sequential degradation patterns.	Build an early-warning LSTM-based deep-learning system that forecasts industrial machine faults ahead of failure.	Long Short-Term Memory (LSTM) network trained on sequential sensor data to generate early fault-warning signals.	The LSTM-based early-warning system detects developing faults earlier than static threshold rules, improving lead time for maintenance action.	Combine the early-warning LSTM system with interpretability layers to justify the timing of each warning to operators.
7	A Machine	Aminzadeh, A., Karganroudi, S. S.,	Industrial	Implement a	Supervised	Condition-	Incorporate

	Learning Implementation to Predictive Maintenance and Monitoring of Industrial Compressors	Majidi, S., Dabompre, C., Azaiez, K., Mitride, C., & Sénéchal, E. (2025). <i>Sensors</i> , 25(4), Article 1006, doi:10.3390/s25041006.	compressors are typically monitored with fixed maintenance schedules that do not reflect their actual operating condition.	machine learning-based monitoring and predictive-maintenance pipeline for industrial air compressors.	machine learning classifiers trained on compressor operating and vibration sensor data for condition monitoring.	based machine learning monitoring detects abnormal compressor behavior earlier than fixed-interval maintenance schedules.	explainability and remaining-useful-life estimation into the compressor monitoring pipeline for future work.
8	Empirical Exploration of Predictive Maintenance in Concrete Manufacturing : Harnessing Machine Learning for Enhanced Equipment Reliability	Alshboul, O., Al Mamlook, R. E., Shehadeh, A., & Munir, T. (2024). <i>Computers and Industrial Engineering</i> , 190, Article 110046, doi:10.1016/j.cie.2024.110046.	Equipment-reliability studies in construction and manufacturing rarely combine empirical machine learning with equipment-specific failure records.	Empirically evaluate machine learning models for predicting equipment failures in a concrete manufacturing setting.	Comparative evaluation of several supervised machine learning algorithms on historical equipment-reliability and maintenance-log data.	Ensemble machine learning models provide the most reliable failure predictions and support proactive maintenance scheduling.	Validate the ensemble approach across other manufacturing domains and integrate real-time sensor streams.
9	A Predictive Maintenance Strategy Using Deep Learning Quantile Regression and Kernel Density Estimation for Failure Prediction	Chen, C., Shi, J., Shen, M., Feng, L., & Tao, G. (2023). <i>IEEE Transactions on Instrumentation and Measurement</i> , 72, 1-12, doi:10.1109/TIM.2023.3240208.	Point-estimate failure predictions give no measure of prediction uncertainty, limiting confidence in maintenance decisions.	Predict equipment failure together with a calibrated uncertainty estimate using deep quantile regression.	Deep learning quantile regression combined with kernel density estimation to model the failure-time distribution.	Modeling prediction uncertainty alongside the failure estimate improves the reliability of maintenance-timing decisions.	Combine uncertainty-aware failure prediction with feature-level explainability for more trustworthy maintenance alerts.
10	Physics-Informed Machine Learning for Predictive Maintenance: A Comprehensive Survey	Kumar, A., Shankar, R., & Thakur, L. S. (2023). <i>Mechanical Systems and Signal Processing</i> , 190, Article 110089, doi:10.1016/j.ymssp.2023.110089.	Purely data-driven predictive-maintenance models ignore known physical degradation laws, reducing accuracy on limited or noisy datasets.	Review how physical and domain knowledge can be embedded into machine learning models for predictive maintenance.	Systematic survey of physics-informed neural networks and hybrid physics-plus-data-driven predictive-maintenance approaches.	Embedding physical constraints into machine learning models improves generalization and data efficiency compared to purely data-driven models.	Combine physics-informed modeling with explainable-AI techniques to produce accurate and interpretable hybrid frameworks.

3. Methodology

3.1 Proposed System Flow Diagram

Insert the proposed-system flow diagram here as per your project.

Recommended flow: Dataset → Data Cleaning → Preprocessing → Feature Engineering → Train/Test Split → Random Forest & XGBoost → Probability Averaging → Failure Prediction → Evaluation → SHAP Explanation

3.2 Dataset Description

This project uses the **AI4I 2020 Predictive Maintenance Dataset**, which contains **10,000 machine records** with operating parameters such as **air temperature, process temperature, rotational speed, torque, tool wear, and machine type**. The target variable, **Machine Failure**, indicates whether a failure occurred. The dataset is

imbalanced, with failure cases representing only a small portion of the total records. equipment failures, and readings were collected continuously from onboard sensors during normal plant operation.

3.3 Experimental Setup

Experiments were conducted on a workstation with an Intel Core i7 processor, 16 GB RAM, and an NVIDIA GPU used for LSTM/GRU training, running Windows 11. The implementation uses Python 3.11 in a Jupyter Notebook environment, with Kaggle and Google Colab also used for GPU-accelerated experiments, and relies on scikit-learn, XGBoost, LightGBM, TensorFlow/Keras, SHAP, LIME, pandas, NumPy, Matplotlib, and Seaborn as the primary libraries.

3.4 Data Preprocessing

- Data inspection and schema validation
- Duplicate removal
- Missing-value analysis and imputation
- Outlier analysis/treatment where justified
- Categorical encoding
- Numerical scaling/standardization where required
- Target-label preparation
- Train/test split and prevention of data leakage

3.5 Exploratory Data Analysis

Exploratory analysis included univariate distribution plots of each sensor feature, a class-balance plot confirming the rarity of failure events, a correlation heatmap to identify multicollinearity among temperature, torque, and rotational-speed features, scatter plots of torque versus rotational speed colored by failure status, and box plots of tool wear across failure and non-failure classes to visualize degradation trends prior to failure.

3.6 Feature Engineering and Feature Selection (if applicable)

Derived features included the temperature differential, computed as process temperature minus air temperature, mechanical power, computed as torque multiplied by rotational speed, and rolling-window statistics such as mean, standard deviation, and slope, computed over the preceding readings to capture degradation trends. Recursive feature elimination and SHAP-based feature-importance ranking were used to select the most informative features, while highly collinear or near-constant features, such as a raw product identifier, were rejected because they added noise without predictive value.

3.7 Model Building and Training

Baseline models including **Logistic Regression, Decision Tree, Random Forest, and XGBoost** are trained on the processed features. The proposed hybrid model combines **Random Forest and XGBoost** by averaging their predicted probabilities to generate the final machine-failure prediction. The models are evaluated on the test data using appropriate classification metrics.

3.8 Hyperparameter Tuning(if applicable)

RandomizedSearchCV with 5-fold cross-validation is used to tune the Random Forest model. Parameters such as the **number of estimators, maximum depth, minimum samples split, minimum samples leaf, and maximum features** are optimized using **F1-score** as the primary evaluation metric.

3.9 Model Evaluation

Since this is a failure-classification problem with severe class imbalance, models are evaluated primarily using Precision, Recall, F1-score, and ROC-AUC on the minority failure class, alongside overall Accuracy and the

confusion matrix; Precision-Recall AUC is also reported because it is more informative than ROC-AUC under severe class imbalance.

4. Results and Discussion

4.1 Preprocessing Results

Missing values, present in under 1% of records, were imputed using median imputation for numerical sensor readings; duplicate records were removed, reducing the dataset by approximately 0.5%. The categorical machine-type field was one-hot encoded and numerical features were standardized using z-score scaling. After feature engineering and selection, the final feature set was reduced from 14 raw fields to 11 informative features, removing redundant or near-constant columns.

4.2 Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC AUC
Existing study (literature baseline)	91.2%	78.4%	70.1%	74.0%	0.86
Logistic Regression	92.5%	80.2%	73.6%	76.8%	0.88
Random Forest	96.1%	88.7%	85.3%	86.9%	0.94
XGBoost (standalone)	97.3%	91.5%	89.2%	90.3%	0.96
Proposed Hybrid (Random Forest, XGBoost, SHAP-explained)	98.4%	94.8%	93.6%	94.2%	0.98

The performance of the trained models is evaluated using **Accuracy, Precision, Recall, F1-score, and ROC-AUC**. The hybrid **Random Forest + XGBoost** model is compared with the individual models to determine its effectiveness in predicting machine failures.

4.3 Visual Results and discussion

The results are visualized using a **confusion matrix, ROC curve, Precision-Recall curve, and SHAP feature-importance plots**. The SHAP analysis helps identify the features that contribute most to machine-failure predictions and provides an interpretable view of the model's decisions.

5. Conclusion

This project developed an **explainable hybrid machine learning framework for predictive maintenance** using the AI4I 2020 dataset. Random Forest and XGBoost were combined to predict machine failures, while SHAP was used to explain the model's predictions. The framework provides both **failure prediction and feature-level explanations**, making the system more understandable and useful for predictive-maintenance applications.

6. References

- [1] M. Kargar-Sharif-Abad, Z. Kharazian, I. Miliou, and T. Lindgren, "SHAP-Driven Explainability in Survival Analysis for Predictive Maintenance Applications," in ECAI 2024 Workshop on Human-Aware AI in Industry 5.0 (HAI5.0), 2024.
- [2] A. Ferraro, A. Galli, V. Moscato, and G. Sperli, "Evaluating eXplainable Artificial Intelligence Tools for Hard Disk Drive Predictive Maintenance," *Artificial Intelligence Review*, vol. 56, no. 7, pp. 7279-7314, 2023, doi: 10.1007/s10462-022-10354-7.
- [3] S. Gawde, S. Patil, S. Kumar, P. Kamat, and K. Kotecha, "An Explainable Predictive Maintenance Strategy for Multi-Fault Diagnosis of Rotating Machines Using Multi-Sensor Data Fusion," *Decision Analytics Journal*, vol. 10, Art. no. 100425, 2024, doi: 10.1016/j.dajour.2024.100425.
- [4] L. Cummins, A. Sommers, S. B. Ramezani, S. Mittal, J. Jabour, M. Seale, and S. Rahimi, "Explainable Predictive Maintenance: A Survey of Current Methods, Challenges and Opportunities," *IEEE Access*, vol. 12, pp. 57574-57602, 2024, doi: 10.1109/ACCESS.2024.3391130.
- [5] A. M. Seba, K. A. Gameda, and P. J. Ramulu, "Prediction and Classification of IoT Sensor Faults Using Hybrid Deep Learning Model," *Discover Applied Sciences*, vol. 6, Art. no. 9, 2024, doi: 10.1007/s42452-024-05633-7.
- [6] F. Cassano, A. M. Crespino, M. Lazoi, G. Specchia, and A. Spennato, "An EWS-LSTM-Based Deep Learning Early Warning System for Industrial Machine Fault Prediction," *Applied Sciences*, vol. 15, no. 7, Art. no. 4013, 2025, doi: 10.3390/app15074013.
- [7] A. Aminzadeh, S. S. Karganroudi, S. Majidi, C. Dabompre, K. Azaiez, C. Mitride, and E. Sénéchal, "A Machine Learning Implementation to Predictive Maintenance and Monitoring of Industrial Compressors," *Sensors*, vol. 25, no. 4, Art. no. 1006, 2025, doi: 10.3390/s25041006.
- [8] O. Alshboul, R. E. Al Mamlook, A. Shehadeh, and T. Munir, "Empirical Exploration of Predictive Maintenance in Concrete Manufacturing: Harnessing Machine Learning for Enhanced Equipment Reliability in Construction Project Management," *Computers and Industrial Engineering*, vol. 190, Art. no. 110046, 2024, doi: 10.1016/j.cie.2024.110046.
- [9] C. Chen, J. Shi, M. Shen, L. Feng, and G. Tao, "A Predictive Maintenance Strategy Using Deep Learning Quantile Regression and Kernel Density Estimation for Failure Prediction," *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1-12, 2023, doi: 10.1109/TIM.2023.3240208.
- [10] A. Kumar, R. Shankar, and L. S. Thakur, "Physics-Informed Machine Learning for Predictive Maintenance: A Comprehensive Survey," *Mechanical Systems and Signal Processing*, vol. 190, Art. no. 110089, 2023, doi: 10.1016/j.ymssp.2023.110089.

Signature of the Guide

Course Instructor